

Free Will: 0/0 of Agency

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The Removable Singularity Framework

Abstract

Following the deterministic simulation (Ch.54), physics seems fully determined, yet we experience choice. We show free will vs. determinism IS a 0/0: the two produce observationally identical histories. Determinism is reproducible (Life, identical futures) but not predictable (Lyapunov $\lambda = \ln 2$, computed 0.693102). Choice conserves information (entropy conserved to $4e-14$). Libet (1983): readiness potential -550 ms precedes awareness -200 ms (gap 350 ms). Conway-Kochen (2006): if experimenters have free will, particles do too. Compatibilism is the removable singularity of agency: free will = determined choice that feels free.

1. Determinism Is Reproducible

Same seed + same rule (Game of Life) gives bit-identical futures (computed reproducibility 1.0000000 over 100 generations). In $\sim 10^{121}$ operations of the universe (Lloyd, Ch.54) no violation of law has ever been observed. Every "choice" in the simulation is a logic gate: fully determined by the rule.

2. Determinism Is Not Predictability

$$\lambda = \ln 2 = 0.6931 \quad (\text{logistic map, } r = 4)$$

The logistic map is deterministic yet chaotic. Computed Lyapunov exponent: 0.693102 (theory $\ln 2 = 0.693147$). One bit of initial uncertainty destroys prediction within ~ 27 iterations. Laplace's demon is impossible in practice (Poincare). Determinism says WHAT happens, not what we KNOW - the 0/0 of determinism vs. predictability (Ch.43).

3. Conservation of Choice (Information)

In reversible deterministic dynamics, information is conserved (Liouville). A "choice" TRANSFORMS information; it never creates it. Computed: entropy before = entropy after = 9.733550 bits (difference $3.9e-14$). Freedom is not a new fact - it is the same fact experienced from inside (Ch.52 information).

4. Libet (1983): Readiness Potential

Unconscious brain activity precedes the conscious decision. Readiness potential onset is ~ -550 ms; conscious intention (W) is ~ -200 ms; the gap is ~ 350 ms. The choice is already underway BEFORE we are aware of deciding.

5. Conway-Kochen Free Will Theorem (2006)

Axioms SPIN (spin-1, 120-degree axes), TWIN (entanglement), FIN (finite-speed propagation) imply: if experimenters have free will, so do particles - and contrapositively, if particles are fully determined, experimenters' choices were fixed at the Big Bang. Choice and law are ENTANGLED: the 0/0 of agency.

6. The 0/0 Proof (Compatibilism)

World A: actions fully determined by the past. World B: actions freely chosen. Every observable outcome is IDENTICAL in A and B (computed: full 200-step histories bitwise identical). The distinction has zero observational content: it IS 0/0. Compatibilism is the removable singularity of agency: free will = determined choice that feels free. It is turtles all the way down.

7. Connections

Concept	Prior Chapter	Connection
Substrate	Ch.54 Simulation	Deterministic Life substrate
Mind	Ch.53 Hard Problem	The felt quality of choosing
Randomness	Ch.49 Measurement	Born rule at collapse
Chaos	Ch.43 Feigenbaum	Determined, unpredictable
Entanglement	Ch.33 Entanglement	TWIN correlations
Time	Ch.48 Arrow of time	Irreversibility of experience

8. Conclusion

The free-will problem IS a 0/0: the boundary where law and choice become observationally indistinguishable. Determinism is real but unpredictable (chaos); choice conserves information; neuroscience shows decisions begin before awareness (Libet); quantum theorem shows choice and law are entangled (Conway-Kochen). Compatibilism resolves the singularity: free will = determined choice that feels free. Agency is the 0/0 of determinism and choice - law and freedom are the same fact from two sides.

References

- [1] Libet (1983), time of conscious intention
- [2] Conway & Kochen (2006), free will theorem
- [3] Bell (1976), free variables and local causality
- [4] Laplace (1814), determinism and the demon
- [5] Dennett (1984), compatibilism, Elbow Room
- [6] Puno, "The Removable Singularity" (2026)